

Enabling fundamental understanding of Nature with novel binning methods for 2D histograms

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ABSTRACT

Context. Visualization of 2D distributions is an essential task, commonly done with a 2D histogram. The histogram is built by subdividing the sample space into regions and color-coding the number of samples in each region.

Aims. We aim to solve long-standing problems with common 2D histogram methods: lack of thematic, visual, and conceptual unity with underlying data, and general stagnation in the field.

Methods. We develop a new method for plotting 2D histograms with arbitrary bin shapes, including aperiodic tilings and geographic maps. We apply the method to several common plot types from the literature.

Results. We find our method performs best across all tasks, solving the problems and propelling the scientific progress forward.

1. Introduction

Visualizing two-dimensional distribution samples is essential for reporting any modern scientific result. The main tool for doing that is the 2D histogram, constructed by dividing the sample space into regions (bins) and using color to show the number or density of samples in a given bin. Yet, despite the immense importance of 2D histograms, for a long time, data visualization practitioners have faced debilitating stagnation. Most tools (e.g. Matplotlib Hunter (2007), ROOT Brun and Rademakers (1997)) limit them to simple rectangular or, at best, hexagonal bins. Some domain-specific binning methods, such as HEALPix Górski et al. (2005), exist, but their applications are naturally limited.

To solve this long-standing issue in scientific and broader data analysis communities, we propose a novel method: **Fundamental Understanding of Nature Binning**, abbreviated as **funbin**. It allows one to produce 2D histograms using a wide variety of bin shapes, including but not limited to periodic and aperiodic plane tilings, Voronoi diagrams, outlines of geographical areas, and others. The paper is structured as follows: in Section 2 we outline the conceptual foundations of **funbin**, drawing on modern and ancient philosophical and mystical traditions; in Section 3 we describe a reference **funbin** implementation as Matplotlib-compatible Python package; in Section 4 we illustrate the depth and breadth of scientific breakthroughs to be brought about by **funbin** with a few examples from astro- and particle physics; finally, we conclude in Section 5.

2. Conceptual foundation

Hermes Trismegistus is an ancient mystical and philosophical figure who features characteristics of both Hermes and Thoth. The former is a Greek god of travelers, merchants, and orators. The latter is an Egyptian god of, among others, wisdom and learning. It is therefore not surprising that a modern scientist, struggling for knowledge, but also precariously searching to sell

this knowledge, can find inspiration in the writings of Hermes Trismegistus.

To outline the conceptual foundation of **funbin**, we draw on one of the most famous texts by the "thrice great", the Emerald Tablet Trismegistus (-100). In turn, one of the most famous quotes from the text says (the translation from Holmyard (1923)):

That which is above is from that which is below, and that which is below is from that which is above, working the miracles of one [thing]. As all things were from One.

In medieval and Renaissance times, this principle was adopted to state the similarity between the Microcosm and Macrocosm, for example, in Paracelsus (1567). It is now our turn, not dissimilar to Paracelsus, to base our practices on near-inscrutable writing from centuries long gone. In the context of this work, the Macrocosm represents the objects under study in the world. The Microcosm is, in turn, the interior region of a scientific work, and most importantly, the plots. Therefore, the adaptation of Hermeto-Paracelsian insight to scientific practice is to find visual, conceptual, and thematic parallels between objects under study and the way one decides to visualize them.

If we examine the types of distribution one usually aims to visualize, we readily find almost none of them to look like squares. Nature rarely produces C_4 -symmetric shapes, instead favoring fuzzy blobs. Therefore, the widespread use of square or rectangular bins comes into pronounced conflict with the vast majority of the Macrocosm. One might argue that hexagonal bins improve on the situation by being closer to a circular shape (continuous circular symmetry) while still tiling the plane. However, in our view, replacing unnatural C_4 symmetry with a slightly more complex C_6 is merely a band-aid solution.

Moreover, both rectangular and hexagonal bins tile the plane in a periodic way. More precisely, both tilings feature global translational symmetry, that is, some translations map the tiling onto itself. This fact alone sets scientific research up for failure: just like the grid one uses, one is forced to indefinitely repeat the same pattern.

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Finally, even if one manages to escape the trap of endless repetition and the pitfalls of squaring non-squares, the ubiquity of conventional binnings leads to the lack of conceptual resonance between the Micro- and Macrocosm. We might imagine rare cases where this is not true: a lattice QCD researcher using square binning to mirror the spacetime discretization, or a melitologist binning their data in hexagonal bins to evoke an image of the honeycomb. However, in most cases, the conventional binnings do not evoke any image in the Macrocosm of the work.

3. Method

We have implemented the principles of `funbin` in an eponymous Python package¹. The package comprises two main parts: the main plotting function, also called `funbin`, and a suite of algorithms to generate polygon sets for various binnings. Below is a self-contained example script:

```
import numpy as np
from numpy.random import normal, random
from matplotlib import pyplot as plt
from funbin import funbin
from funbin.einstein import aperiodic_monotile

# mixing 80/20 between two offset Normal modes
n = 50000
samples = np.where(
    random(n) > 0.2,
    normal(0, 1.0, (2, n)),
    normal([[2.0], [2.0]], 0.5, (2, n))
)

# tiling is simply a list of Polygon objects
tiling = aperiodic_monotile(bins=(40, 40))

# API largely mirrors ax.hexbin
fig, ax = plt.subplots()
funbin(ax, *samples, tiling=tiling)
ax.set_aspect("equal")
fig.savefig("funbin-example.png")
```

The function `funbin` admits an arbitrary set of polygons to be used as bins. The algorithm is straightforward:

1. Translate and stretch polygons to match the data bounding box.
2. For every data point, find a polygon it belongs to. This is done with the help of an auxiliary "spatial index" that rejects far-away polygons without running the point-in-polygon algorithm.
3. Compute sample weight sum for each polygon. For an un-weighted histogram, all weights are 1, so the weight sum is a simple count.
4. Divide each polygon's weight sum by its area to estimate sample weight density within its boundaries.
5. Plot polygons with their colors defined by the sample weight density and a colormap.

Algorithms to generate polygon sets include:

- Penrose tilings P1, P2, P3 Penrose (1979)
- Aperiodic monotile Smith et al. (2024)
- Voronoi diagrams Voronoi (1908) of a random or user-specified set of points
- Map data imported from GeoJSON format Butler et al. (2016).

¹ <https://github.com/nj-vs-vh/funbin>

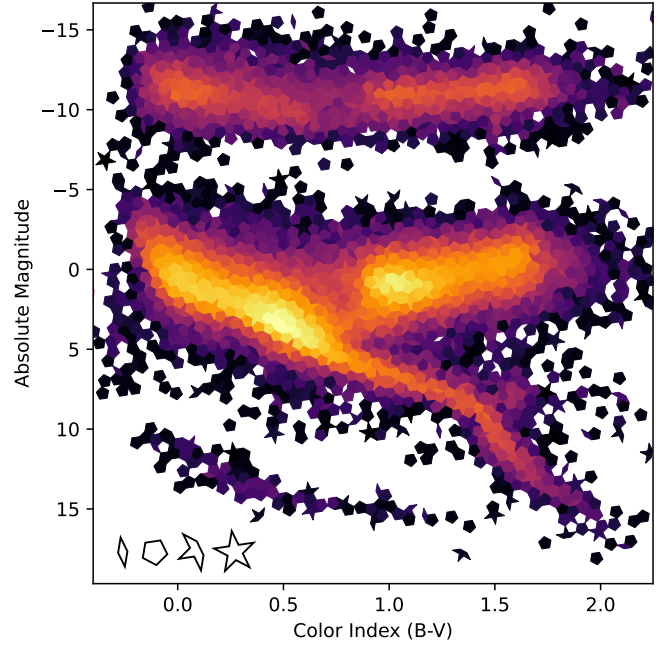


Fig. 1: Hertzprung-Russell diagram binned with Penrose P1 tiling. Data from HYG (Hipparcos, Yale, Gliese) Stellar database is used. Bottom left corner shows tile shapes.

4. Examples

In this section, we apply `funbin` to several example problems and highlight how binning features (Microcosm) enhance one's fundamental understanding of the problem at hand and Nature as a whole (Macrocosm). We first focus on two aperiodic tilings: Penrose tiling of three types and a recently discovered aperiodic monotile with two variants on the tile shape. Their aperiodicity itself automatically elevates the scientific value of the plotted data as it breaks the malignant repetitiveness associated with conventional periodic tilings, described in Section 2. For each tiling, we find a particular dataset that is especially concordant with its shape. Finally, we explicitly explore the Microcosm-Macrocosm resonance by using Earth's geographical data to map out stellar mass distribution in the Universe.

4.1. Penrose tilings

Penrose tilings are a set of aperiodic tilings described in Penrose (1979). The first of them chronologically, P1 tiling consists of five shapes: three kinds of pentagons (they look the same but have different matching rules, so are considered different tiles), a star, a diamond, and a "boat". The imagery is hard to miss: aside from a literal star, the boat is essentially a broken star, perhaps disrupted by tidal forces or collision; the pentagon is nothing but a star with no rays coming out from it, an extinct star, with multiplicity of similar pentagons showing distinct endpoints of stellar evolution: black hole, neutron star, white dwarf; finally, the diamond can be interpreted as referring to the crystalline structure at the cores of carbon-rich white dwarfs. Driven by these considerations, in Figure 1 we plot data from HYG database D. as a binned Hertzprung-Russell diagram, using Penrose P1 tiling as bins.

The most widely known Penrose tiling is P3, consisting of two types of rhombus, sometimes dubbed "fat" and "thin". An-

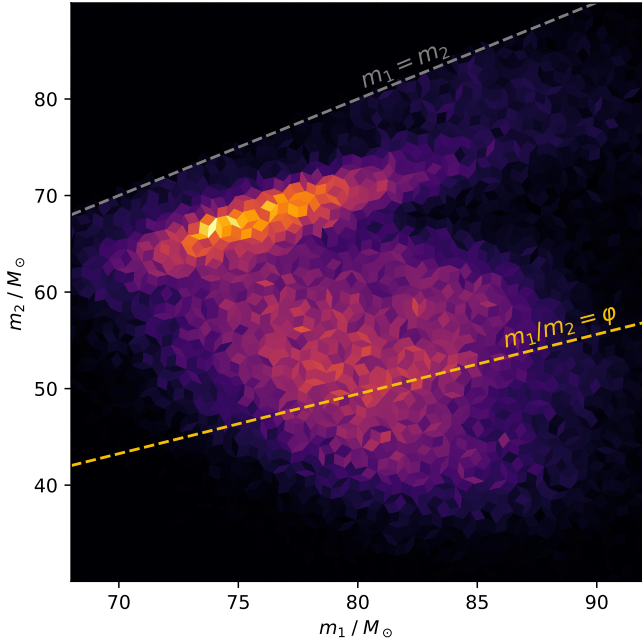


Fig. 2: Posterior distribution of black hole masses from GW191109_010717 binned with Penrose P2 tiling.

other situation where two similar but somewhat differently-sized objects come together to rather spectacular results is a black hole merger. It is not surprising, then, that Roger Penrose himself has contributed greatly to the study of black holes. In the last ten years, great progress has been achieved in observing binary black hole mergers through gravitational wave interferometry Abbott et al. (2016). Gravitational wave astronomy has also been one of the leading fields in applying Bayesian statistics for event reconstruction, which in practice implies working with multidimensional samples from the posterior distribution of interest. All this makes GW astronomy another field to be readily revolutionized by *funbin*.

To illustrate this, in Figure 2 we plot a posterior distribution of two black holes' masses (marginalized over all other parameters) from the event GW191109_010717 published by the LVK collaboration as part of the GWTC-3 parameter estimation data release LVK Collaboration (2023). We notice that the two types of rhombus used in the tiling, combined with gradient shading from the posterior distribution, create structures resembling projections of 3D cubes. This emergent property highlights the power of *funbin* to show higher-dimensional structures in the data, even when technically it is limited to two. Moreover, we notice that a secondary mode of the posterior features black holes with a mass ratio equal to the golden ratio $\varphi \approx 1.618$, shown in a golden dashed line. Even if this is not remarkable on its own, when paired with the fact that the golden ratio also gives the ratio between areas of fat and thin rhombi, this clearly shows the ability of *funbin* to highlight otherwise obscure mathematical relations in the data it visualizes.

Finally, the Penrose P2 tiling uses two types of tiles, sometimes dubbed "darts" and "kites". Both darts and kites are designed to fly, and darts are specifically designed to precisely hit a target when thrown from some distance. Perhaps the most impressive human-made dart-throwing device is the Large Hadron Collider with multi-TeV proton darts. Also, the darts and kites can be reinterpreted as symbolic representations of particle jets

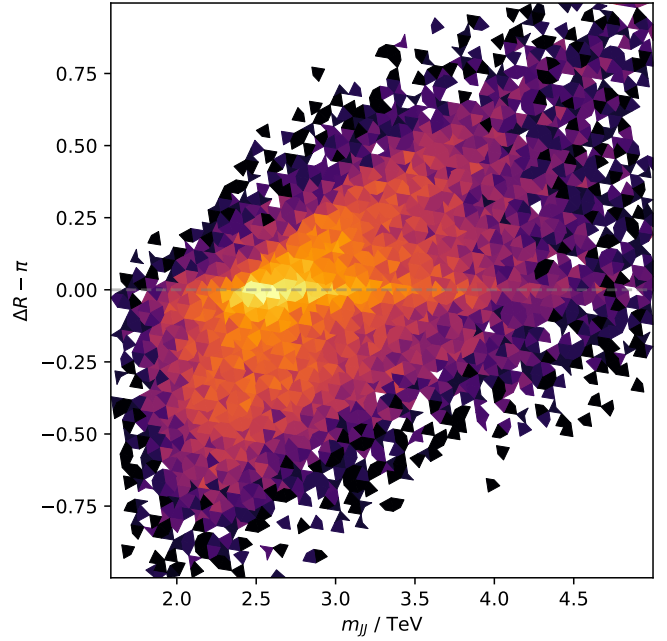


Fig. 3: Total invariant mass of two highest jets m_{JJ} versus their η - ϕ cylinder distance in LHC2020 BlackBox1 dataset, binned with Penrose P3 tiling.

with different substructure: larger intensity in the core (kite) or in the shell (dart) of the jet. For these reasons, in Figure 3 we use LHC simulation data from LHC Olympics BlackBox1 dataset Kasieczka et al. (2019) with Penrose P2 binning. We use FastJet Cacciari et al. (2012) to analyze simulated jets and plot the 2D histogram of the invariant mass of the main dijet and cylinder (η - ϕ) distance between its two jets. The intent behind the LHC2020 dataset is to provide a benchmark for anomaly detection algorithms. However, we note that by using a non-standard binning and a potentially buggy implementation *funbin* allows one to instead inject new anomalies into the data, opening exciting new avenues for high-energy physics.

4.2. Aperiodic monotile

As explained in Section 4.1, aperiodicity itself is a highly desirable property for binnings used in scientific plots. Until recently, only two-tile aperiodic tiling, such as P2 and P3, have been known. But in 2023, a single shape that can tile the whole plane with no repeats, an aperiodic monotile, was discovered Smith et al. (2024). It is also called einstein, from "ein Stein", meaning "one stone" in German. The addition of singularity to aperiodicity has profound implications for the Hermeto-Paracelsian reasoning: the tiling is built on a single principle, yet it never repeats itself. Thus, it provides a geometrical foundation for the scientific method, which is also a singular principle, yet it is supposed to facilitate an endless non-repeating procession of scientific activity.

Inspired by meditations on the nature of (a)periodicity, we turn our attention to pulsars. Pulsars are highly magnetized and fast-rotating neutron stars, and they constitute probably the most extreme case of periodic behavior in the Universe. At the same time, they generally show measurable change in their period over time, slowly radiating away the rotational energy, therefore their true nature is aperiodic. For our visualization, we retrieve a sam-

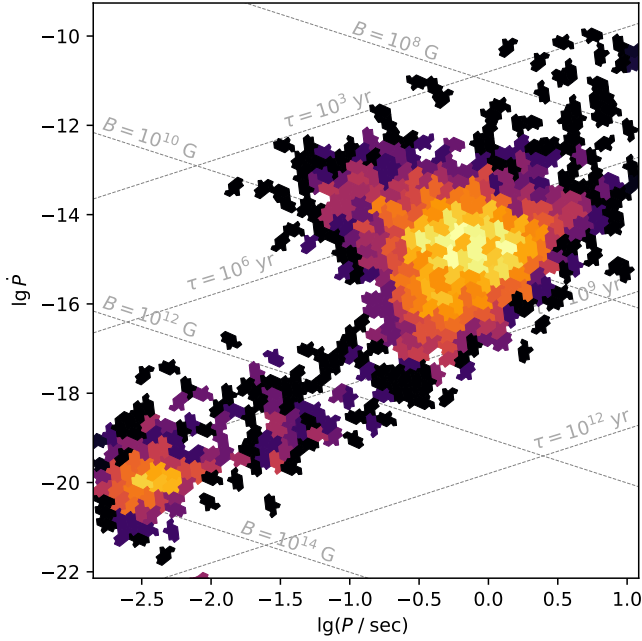


Fig. 4: $P - \dot{P}$ diagram for pulsars from ATNF Catalog Manchester et al. (2005) binned with aperiodic monotile "turtle". Lines of negative slope show constant characteristic lifetime $\tau \equiv \frac{P}{2\dot{P}}$ and lines of positive slope show constant characteristic magnetic field $B \equiv \sqrt{\frac{P}{\sec}} \dot{P} 3.2 \times 10^{19} \text{ G}$

ple of 2798 pulsars from ATNF Pulsar Catalog Manchester et al. (2005)² and reproduce the classic $P - \dot{P}$ diagram in Figure 4. For binning, we use the "turtle" ($\text{Tile}(\sqrt{3}, 1)$) shape from the aperiodic monotile family. The name calls to mind the fable of the tortoise (turtle's relative) and the hare: like the hare, the pulsar is very fast, but is recklessly slowing down and sooner or later (in $\tau = 10^3$ to 10^6 years) risks being overtaken even by a turtle. Aside from profound scientific insights, funbin is capable of life advice.

In another pulsar-related application, in Figure 5 we use a different aperiodic monotile from the same family, "hat" ($\text{Tile}(1, \sqrt{3})$), as a pixel for γ -ray data from Fermi LAT Atwood et al. (2009). Namely, we have selected 0.2 – 10 GeV photons in a 22° radius around the direction of PSR J2032+4127 (RA = 308.055, Dec = 41.4568) recorded during the year 2025. The data show slowly varying diffuse emission and three bright sources, pulsars J2021+4026, J2021+3651, and J2032+4127. The bottom right corner shows the Fermi LAT point-spread function around 1 GeV energy if it were shaped like the hat tile.

4.3. World map

In this section, we achieve an explicit Microcosm/Macrocosm convergence by bringing together two spheres: Earth's political map provides binning for stellar mass distribution across the local Universe within 200 Mpc from Biteau (2021). For the former, we use the Natural Earth dataset Earth and volunteers to extract administrative boundaries of all countries at 110 m precision. Since Earth's oceans do not have administrative boundaries, they would constitute a single giant bin, which is inspiring theoretic-

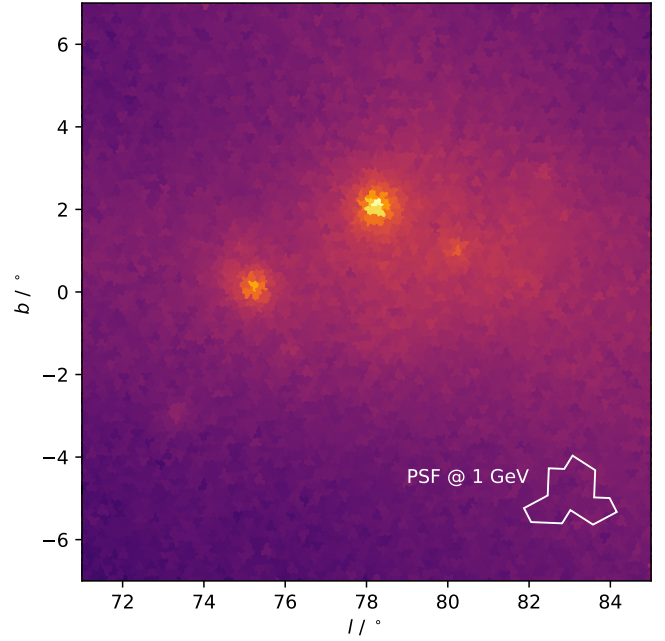


Fig. 5: Raw photon γ -ray counts detected by Fermi LAT in the sky area around PSR J2032+4127, binned with aperiodic monotile "hat".

cally, but leads to a poorly readable plot. Therefore, we subdivide the ocean with the regular HEALPix bins ($n_{\text{side}} = 8$). For the data we extract from the Biteau (2021) galaxies with $d_L < 200$ Mpc and weight them according to their estimated M_* , accounting for the mass incompleteness correction. We then map their Galactic longitude and latitude to Earth's longitude and latitude and juxtapose them on our binning. The resulting 2D histogram in Mollweide projection is shown in Figure 6.

funbin allows us to draw profound conceptual conclusions here. By juxtaposing the spheres of Earth and Universe, we essentially rediscover Kantian epistemology: the Earth's map serves as a region of transcendental, a frame structuring our experience of the Cosmos. Perhaps there is already a post-Kantian, even Marxist twist here: the transcendental appears to be historically conditioned and produced. This leads to a speculative connection of funbin with politics. For example, if one desires the plot to have greater angular resolution in the region of the Virgo cluster, one is naturally drawn to the political goal of breaking a large pixel of modern-day Russia down into smaller ones. One could also make the case for abolishing the Antarctic Treaty and dividing Antarctica into countries to gain resolution in the South Galactic Pole region.

We conclude this brief exploration with another famous and relevant quote from Kant (1999):

Two things fill the mind with ever-increasing wonder and awe, the more often and the more intensely the mind of thought is drawn to them: The starry heavens above me and the moral law within me.

It is regrettable, but not unexpected, that the role of the moral law in this analogy is played by the borders of nation-states.

² <https://www.atnf.csiro.au/research/pulsar/psrcat/>

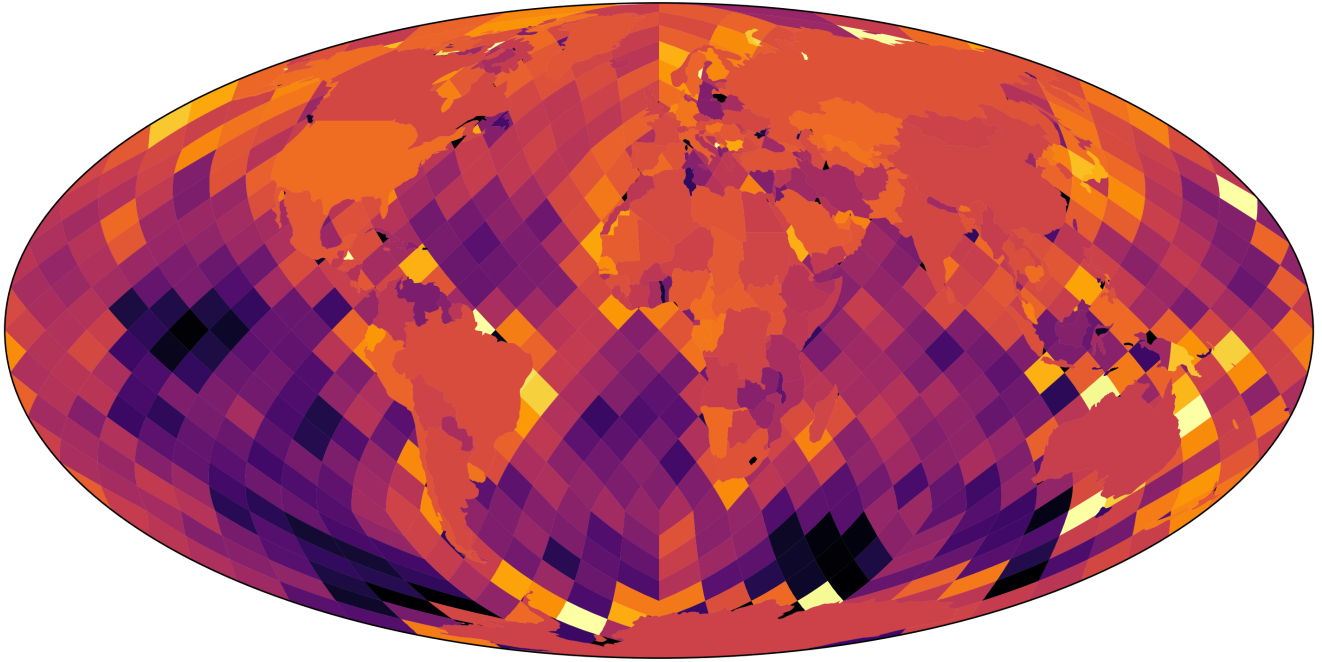


Fig. 6: Galaxies in the local Universe ($d_L < 200$ Mpc) from Biteau (2021), weighted by their stellar mass and shown in Galactic coordinates, binned with Earth's political map and HEALPix subdivision of the oceans.

5. Discussion

We have presented our solution to long-term problems in the field of data visualization, namely concerning 2D histogram plots: **Fundamental Understanding of Nature Binning**, or **funbin**. We have outlined multiple issues with existing bin shapes (rectangular and hexagonal), namely, a lack of visual, thematic, and philosophical harmony between them and the data they are supposed to represent. We draw on ancient, medieval, and early modern mysticism and philosophy to create a Hermeto-Paracelsio-Kantian framework, which we apply to several data visualization tasks from astro- and particle physics. We find that **funbin** solves a significant portion of the problems (while also introducing interesting new ones) thus constituting a significant step in the scientific progress.

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